WORKED SOLUTION

STEP II 2015, Question 1

Pure • Inequalities and Series

STEP II 2015, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q1

The key idea

Each inequality between a function and a logarithm becomes, once we substitute a well-chosen value of x, an inequality between a single term of the series and a *difference* of logarithms. Summing those differences makes almost everything cancel (the sum telescopes), leaving one or two logarithms. The art is in choosing the substitution so that the logarithm splits into a difference.

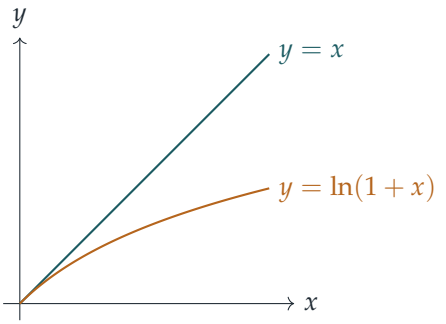
Solution

Part (i)

Let $f(x) = x - \ln(1+x)$ for $x \geq 0$. Then $f(0) = 0 - \ln 1 = 0$ and

$$f'(x) = 1 - \frac{1}{1+x} = \frac{x}{1+x} > 0 \qquad \text{for } x > 0.$$

So f is strictly increasing on $[0, \infty)$, and for every $x > 0$ we have $f(x) > f(0) = 0$. That is, $x > \ln(1+x)$ for all positive x, as required.



Now take $x = \frac{1}{k}$ for a positive integer k:

$$\frac{1}{k} > \ln\left(1 + \frac{1}{k}\right) = \ln\left(\frac{k+1}{k}\right) = \ln(k+1) - \ln k.$$

Adding these for $k = 1, 2, \ldots, n$, the right-hand sides telescope:

$$\begin{aligned}\sum_{k=1}^{n} \frac{1}{k} &> (\ln 2 - \ln 1) + (\ln 3 - \ln 2) + \cdots + (\ln(n+1) - \ln n) \\ &= \ln(n+1) - \ln 1 = \ln(n+1).\end{aligned}$$

This is the required result. (As a by-product, the harmonic series diverges, since $\ln(n+1) \to \infty$.)

Part (ii)

Let $g(x) = x + \ln(1-x)$ for $0 \le x < 1$. Then $g(0) = 0$ and

$$g'(x) = 1 - \frac{1}{1-x} = \frac{-x}{1-x} < 0 \qquad \text{for } 0 < x < 1,$$

since $1 - x > 0$ there. So g is strictly decreasing and $g(x) < 0$ for $0 < x < 1$; that is,

$$x < -\ln(1-x) \qquad (0 < x < 1).$$

Ask yourself What value of x should we substitute, and why can't we use $k = 1$?

We want $\frac{1}{k^2}$ on the left, so take $x = \frac{1}{k^2}$. This needs $x < 1$, so $k \ge 2$; for $k = 1$ the logarithm $\ln(1-1)$ is undefined. That is why the term $\frac{1}{1} = 1$ is set aside, and it is exactly the 1 in the bound $1 + \ln 2$. For $k \ge 2$, the key is that $1 - \frac{1}{k^2}$ factorises:

$$1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2},$$

so

$$\frac{1}{k^2} < -\ln\frac{(k-1)(k+1)}{k^2} = (\ln k - \ln(k-1)) - (\ln(k+1) - \ln k).$$

Write $a_k = \ln k - \ln(k-1)$, so the right-hand side is $a_k - a_{k+1}$. Summing from $k = 2$ to N telescopes:

$$\sum_{k=2}^{N} \frac{1}{k^2} < a_2 - a_{N+1} = \ln 2 - \ln\left(\frac{N+1}{N}\right) < \ln 2,$$

because $\ln\frac{N+1}{N} > 0$.

So the partial sums $\sum_{k=1}^{N} \frac{1}{k^2}$ increase with N and are all less than $1 + \ln 2$. An increasing sequence that is bounded above converges, so the infinite series converges, and its sum is at most $1 + \ln 2$.

To make the inequality strict, note that passing to the limit only preserves $\leq$, so we keep a little room. The $k = 2$ inequality, $\frac{1}{4} < \ln\frac{4}{3}$, is strict; let $\delta = \ln\frac{4}{3} - \frac{1}{4} > 0$. Then for every $N \geq 2$ the same telescoping gives

$$\sum_{k=2}^{N} \frac{1}{k^2} < \ln 2 - \ln\left(\frac{N+1}{N}\right) - \delta < \ln 2 - \delta,$$

so in the limit $\sum_{k=2}^{\infty} \frac{1}{k^2} \leq \ln 2 - \delta < \ln 2$. Adding the $k = 1$ term,

$$\sum_{k=1}^{\infty} \frac{1}{k^2} < 1 + \ln 2,$$

which is the required result.

Checks

- Part (i) with $n = 1$: $1 > \ln 2 \approx 0.693$. With $n = 10$: $\sum_{k=1}^{10} \frac{1}{k} \approx 2.929$, while $\ln 11 \approx 2.398$.
- Part (ii): the true sum is $\frac{\pi^2}{6} \approx 1.6449$, comfortably below $1 + \ln 2 \approx 1.6931$. In fact our working gives the slightly sharper bound $1 + \ln 2 - \delta \approx 1.6555$, with $\delta \approx 0.0377$.
- The telescoped bound $\ln 2 - \ln\frac{N+1}{N}$ for $\sum_{k=2}^{N} \frac{1}{k^2}$ was confirmed by computer for every N up to 10^5.

Common slips

- Substituting $x = \frac{1}{k^2}$ for $k = 1$ as well: $\ln(1 - x)$ is then undefined, and forgetting to set the first term aside loses the 1 in $1 + \ln 2$.
- Showing only that $f'(x) > 0$, without $f(0) = 0$: a function can be increasing and still negative.
- Taking limits of a strict inequality and claiming the limit is strict without comment; limits only give $\leq$.

Beyond STEP

Comparing a sum with a logarithm (or an integral) is the start of the integral test, and refining part (i) shows that $\sum_{k=1}^{n} \frac{1}{k} - \ln n$ converges to Euler's constant $\gamma \approx 0.5772$. The exact value $\sum \frac{1}{k^2} = \frac{\pi^2}{6}$ is Euler's solution of the Basel problem, which first-year courses often derive from Fourier series. The monotone convergence argument we used, that bounded increasing sequences converge, is a cornerstone of real analysis; Abbott's *Understanding Analysis* develops it carefully.

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